RESEARCH REPORT · 1 OF 2

What Markets Allow — and What to Build Inside Those Limits

The domain evidence and product design behind a personal command center and learning hub for US equities — NYSE & Nasdaq. What is genuinely knowable, what is folklore, and how to build a daily briefing that respects the difference.

"What is genuinely knowable is risk, not direction. The honest product is a risk-and-context command center: calibrated base rates with confidence intervals, decay stamps on every published pattern, and a firm refusal to play oracle."

— The one-paragraph conclusion of this report

Prepared for Bishant · myStockMarket **Date** July 2026 **Companion volume** Build Blueprint

Contents

PART 1

How to read this report

This document exists to answer one question honestly: *what can a personal market app actually know?* Everything else — the dashboard, the signals, the learning hub — is designed downstream of that answer.

1.1 What this is

This is the first of two volumes. It covers the **domain research** (Parts 2–7): how equity markets move, which analytical techniques have empirical support, how news and events drive prices, and what the record says about retail trading outcomes. It then converts that evidence into **product design** (Parts 8–9): the information architecture, the daily-briefing experience, and the learning system. The second volume — the *Build Blueprint* — covers architecture, the technology stack, data sources with verified pricing, data models, the implementation roadmap, and risks.

Scope: US equities, explicitly. Everything in both volumes is anchored to the US market — NYSE and Nasdaq listings; US session structure (pre-market 4:00–9:30am ET, regular session 9:30am–4:00pm ET, after-hours 4:00–8:00pm ET); the US earnings calendar with its before-open/after-close timing; and US macro catalysts (FOMC, CPI, the jobs report). The evidence in Parts 2–7 is drawn from US data unless a study is explicitly labeled otherwise; the retail statistics include the major non-US censuses (Taiwan, Brazil) precisely because they are the best-evidenced numbers anywhere, applied here to a US-market product. Non-US markets are future extensions, not part of this design. The user is US-based — Long Island, New York — so the product runs on market time (America/New_York) end to end.

1.2 How the research was produced and checked

The findings here were assembled by parallel research passes across six domains (market efficiency, technical-analysis evidence, catalysts, microstructure and retail outcomes, quantitative methods, and machine learning), each required to chase primary sources — peer-reviewed journals, SSRN/NBER working papers, and regulator publications — and to report effect sizes and post-publication decay, not just directions. Eighty-four falsifiable claims were extracted; the twelve most load-bearing were then **adversarially cross-examined** by independent review passes explicitly instructed to refute them. Several were narrowed as a result, and the corrected versions are what appear in this report. Where a correction matters, the text says so plainly (for example, candlestick patterns are reported as *contested-negative*, not "cleanly debunked"). Claims that rest on industry data rather than peer review are labeled as such. What the report is still unsure about is collected in Part 10 rather than hidden.

1.3 The grading language

Every technique in the evidence ledger carries one of four verdicts. They are used consistently across both volumes and — by design — in the app itself:

Verdict	Meaning
SUPPORTED	Replicated in credible out-of-sample tests; the effect is real, though usually smaller than advertised and often hard for an individual to capture after costs.
MIXED	Real statistical content, but contested, decayed, horizon-limited, or captured mainly by institutions. Useful as context; unreliable as a personal edge.
WEAK	Occasional positive findings that fail under data-snooping corrections, out-of-sample tests, or realistic costs.
FOLKLORE	No credible supporting evidence, or the careful evidence points the other way.

STANDING COMMITMENT

Numbers in this report carry their context: sample sizes, confidence intervals, whether the figure is gross or net of costs, and whether the effect decayed after publication. The app this report designs makes the same commitment in its interface — that is the point.

PART 2

How markets actually move

Four facts carry almost all the weight: markets are nearly efficient, returns are violently non-Gaussian, published edges decay, and costs are the only guaranteed prediction. Everything in this product follows from them.

2.1 Nearly efficient — and why "nearly" doesn't help you

The efficient-markets baseline[1] says prices already reflect available information, so price changes come mostly from *new* information — which is unpredictable by definition. The refinement that matters is Grossman & Stiglitz's impossibility result[2]: perfectly efficient markets are impossible, because if prices reflected everything, nobody would be paid to do the reflecting. Markets therefore stay *almost* efficient, with a residual inefficiency just large enough to compensate professionals for the cost of information gathering.

That residual is not free money lying around; it is a **wage** — and it is collected by the people with the lowest costs and fastest information. When a beginner buys a stock because it looks like it will bounce, the counterparty is not another beginner. It is a market-making algorithm or an institutional desk whose effective trading costs are roughly **1.5 basis points** per trade, against a realistic retail cost of **20–57 basis points** once spreads and slippage are counted honestly[3]. The same anomaly can be simultaneously real for them and a losing trade for you. This asymmetry, more than any single study, is why so many "documented" effects in Part 4 are graded *mixed*: real, but not yours to harvest.

2.2 The arithmetic nobody escapes

Sharpe's 1991 argument[4] is close to accounting: the market return is the average of all investors' returns, so after costs the average actively managed dollar must underperform the passive dollar. (Pedersen[5] showed the identity relies on simplifying assumptions — index reconstitution, IPOs and buybacks force even "passive" money to trade — so it is a strong approximation rather than a law.) The empirical record matches the approximation: in the SPIVA scorecards, roughly **93% of all active US equity funds underperformed their benchmark over the trailing 15 years**, and no US category out of 22 shows majority outperformance at that horizon[6]. The figure counts funds equally; weighted by assets the shortfall is smaller (critics estimate nearer 55%[5]) — but the direction is unambiguous.

Design consequence: if full-time professionals with research staffs net-lose to the index, the honest baseline expectation for a beginner picking stocks is underperformance. The app treats a low-cost index position as the default benchmark against which every idea, signal, and paper trade is scored.

Active US equity funds trailing their benchmark, 15 yrs	Average anomaly return lost after publication	Out-of-sample daily-direction hit rate (ML)	Out-of-sample R^2 of volatility forecasts
~93%	**−58%**	**50–55%**	**~0.5**
SPIVA, equal-weighted fund count [6]	McLean & Pontiff, 97 US predictors [8]	vs. "always up" baseline of ~53–55% [24]	HAR/GARCH vs. realized-vol proxies [12]

2.3 Returns are not Gaussian — and your risk display must not be either

Cont's canonical "stylized facts"[7] — confirmed again in 2023–2025 replications on modern data — establish three properties of equity returns: heavy tails, near-zero autocorrelation of returns themselves, and strong autocorrelation of return *magnitudes* (volatility clustering). The practical meaning of heavy tails: under a Gaussian model, the October 1987 one-day crash of −20.5% was a roughly **20–25 sigma** event — probability on the order of 10^{-89}, effectively "cannot happen in the life of the universe." Yet five-plus-sigma days recur every few years. Any interface that draws "expected range" bands from a normal distribution quietly teaches its user that crashes are impossible. The app draws risk bands from fat-tailed or empirical distributions, always.

Volatility clustering has a happier consequence: **the magnitude of moves is genuinely forecastable at short horizons**, even though their direction is not. Daily realized-variance series show first-order autocorrelations around 0.6–0.86, and standard HAR/GARCH models retain out-of-sample R^2 near 0.5 against realized-volatility proxies, one day to one month ahead — with no post-publication decay in nearly two decades, because knowing tomorrow will be turbulent does not tell you which way it breaks[12]. Two honest caveats ride along: forecast power fades beyond roughly 15–20 trading days, and models understate risk around sudden regime breaks. This is the single most important asymmetry in the report: *the app may forecast bumpiness; it may never forecast direction.*

2.4 The decay law: published edges are already half-eaten

McLean & Pontiff tracked 97 published return predictors and found returns **26% lower out-of-sample** (bounding how much was data-mining all along) and **58% lower post-publication**, with the ∼32-point difference attributed to informed traders acting on the paper[8]. Decay is partial, not total — about 40% of the average premium persists — but it concentrates in illiquid, costly-to-trade stocks, exactly where retail execution is worst. Two companion facts sharpen the lesson. First, the multiple-testing problem: with thousands of researchers mining the same data, Harvey, Liu & Zhu argue a newly "discovered" factor should clear a t-statistic of 3.0, not 2.0, before being believed[9]. Second, the replication debate resolved into a both-things-are-true synthesis: most factor *themes* replicate across 93 countries[10], *and* their tradable returns shrink materially after discovery. (A caveat recorded in Part 10: the post-publication decay finding is essentially a US result — it does not reliably replicate across 38 non-US markets[11].)

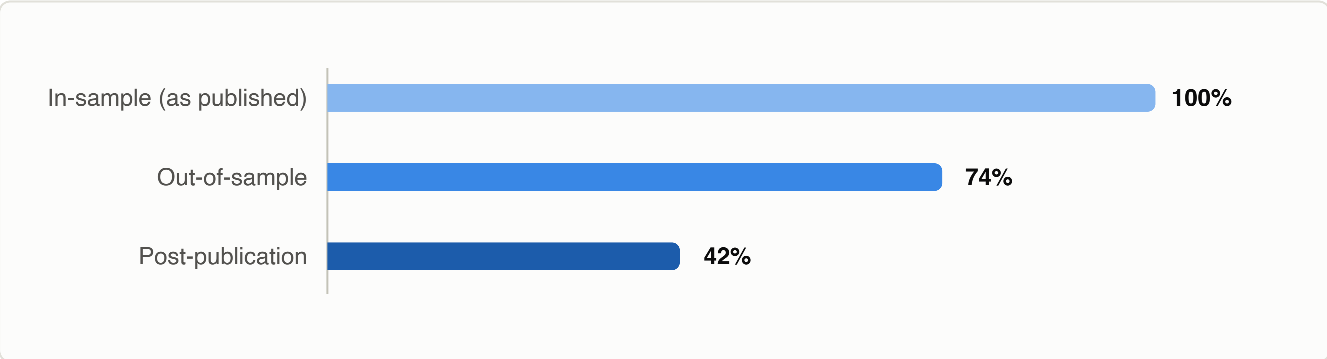

Figure 2.1 — The decay law. Average return of 97 published US stock-market anomalies, relative to their in-sample size: 26% evaporates out-of-sample (it was data-mining), a further 32 points after publication (it was traded away). McLean & Pontiff, Journal of Finance 2016 [8]. The app stamps every published pattern with its publication date and this expected haircut.

2.5 The one perfect forecast: your own costs

Every round-trip trade pays the effective spread plus slippage — a few basis points on mega-caps, several *percent* on small, low-priced names — plus taxes. Multiply by turnover and you have the only prediction in this domain with no error bars: a frequent trader's cost drag is exact arithmetic and guaranteed negative. For the most active retail households in the classic Barber–Odean sample, it summed to about **6.5 percentage points of underperformance per year**[26]. The app makes this arithmetic a first-class display — a "cost mirror" computed from the user's own paper-trading turnover — because it is the most useful honest forecast available to a beginner.

> WHERE THIS LEAVES US
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> Markets are a hostile environment for prediction and a workable environment for **preparation**: knowing what is scheduled, how large moves typically are, how turbulent the next weeks are likely to be, what a pattern's real base rate is, and exactly what trading costs. That reframing — from oracle to preparation — is the product thesis.

PART 3

What is knowable — and what is not

The dividing line below is the most important content in this report. Everything on the left can appear in the app as a computed, confidence-intervalled display. Everything on the right is either excluded or shown only with an explicit "this is not predictable" frame.

3.1 Genuinely knowable

- **Volatility, one day to ~one month out.** Realized-variance persistence is high and HAR/GARCH forecasts hold out-of-sample R^2 near 0.5 against realized-vol proxies, with no post-publication decay in ~two decades[12]. Fades beyond ~15–20 trading days; understates risk at regime breaks. This powers honest "expected range" bands.
- **Fat-tailed crash risk.** A distributional fact at all horizons, immune to costs[7]. Risk displays must be non-Gaussian.
- **Monthly-horizon factor tilts** (value, momentum, quality, low-vol) — real across 93 countries[10], but expect a 30–60% haircut from published numbers[8], and an individual realistically captures them through low-cost factor funds, not a personal scanner.
- **The likely *size* of an event move.** The options-implied straddle is a roughly unbiased ex-ante estimate of earnings-day move magnitude[39] — honest context for "what to watch," with no claim about direction.
- **Your own cost drag.** Effective spread × turnover, plus slippage and taxes, is exact arithmetic — the single most reliable forecast the app can ship, and it is a guaranteed negative[3].
- **Execution timing mechanics.** Spreads are widest in the first ~30 minutes (>20bp on S&P 500 names at the open vs <5bp late day); the closing auction is the deepest liquidity window of the US trading day (industry-measured figures)[66]. Free, capturable knowledge.
- **Base rates of retail outcomes.** Census-grade statistics (Part 6) — displayable as facts, not opinions.
- **The decay law itself.** Published US anomalies lose ~26% out-of-sample, ~58% post-publication on average[8] — so every "proven strategy" can be stamped with an expected haircut.

3.2 Not knowable (by this app, or by anyone selling you a course)

- **Next-day direction of a stock or index, net of costs.** Methodologically sound out-of-sample tests cluster at 50–55% hit rates on modern liquid markets — against a naive "always up" baseline of ~53–55%[24]. Pre-2010 edges existed and decayed after costs[69], so this is an empirical regularity about today's markets, not a theorem — but no persistent, cost-surviving retail edge is demonstrated today. Published 60%+ accuracies typically reflect leakage or denoising artifacts.
- **Market timing from valuation dials.** Dividend yields, P/E ratios and rate spreads fail to beat the naive historical-mean forecast out-of-sample; over a third of newer predictors lost even in-sample significance on re-testing[35].
- **The direction of an earnings, FOMC, or macro reaction.** Implied moves forecast size, not sign; guidance routinely dominates the EPS print[41].

- **Which regime comes next.** Calm/crisis regimes are identifiable after the fact; real-time regime timing has weak out-of-sample support.
- **Why the market moved today** — often. Half to two-thirds of large daily moves lack a cleanly identifiable news cause[55,56]; most morning-after narratives are invented. The app must be willing to say "no clear reason."
- **Whether a small-sample pattern is real.** 60 wins in 100 has a 95% confidence interval of roughly [50%, 69%] — indistinguishable from a coin flip at realistic edge sizes. About 1,000 independent observations are needed for a ±3-point interval.
- **A teachable path to day-trading profitability.** Full-population data show no learning-by-experience (Brazil, 300+ sessions[58]); the tiny profitable minority reflects rare pre-existing skill (Taiwan[57]).
- **Calibrated probabilities from a language model.** Verbalized LLM confidences are systematically overconfident across model families[74] — they are suppressed in this product, never displayed as market probabilities.
- **Institutional edges at retail costs.** Effects that are real at 1.5bp institutional execution do not transfer to 20–57bp retail costs[3]. "The anomaly survives costs" conclusions do not apply to a scanner user.

DESIGN CONSEQUENCE

The app is a **risk-and-context engine with a memory**, not a prediction engine. It forecasts volatility bands and event magnitudes, computes base rates with confidence intervals and decay stamps, explains catalysts with citations, refuses direction, and keeps an append-only public record of how its own flags resolved. Anything that would require the right-hand list is out of scope — permanently.

PART 4

The evidence ledger

Eighteen techniques a beginner will encounter in the first month of reading about trading, graded against the peer-reviewed record. "Retail capture" asks the question that matters: even if real, can an individual harvest it after costs?

LEDGER — TECHNICAL & QUANTITATIVE TECHNIQUES, AS OF JULY 2026

Technique	Verdict	What the careful evidence shows	Retail capture?	Refs
Cross-sectional momentum (3–12 mo winners)	SUPPORTED	~1%/month gross long-short historically; replicates across eight asset classes and two centuries. Apply the ~58% post-publication haircut. Suffers rare multi-month crashes: −73% (Mar–May 2009), −91% (Jul–Aug 1932), gross of costs.	Via low-cost momentum funds; not via a personal scanner	[13,14]
Volatility forecasting (GARCH/HAR)	SUPPORTED	The robust one: OOS $R^2 \approx 0.5$ vs realized-vol proxies at 1-day–1-month horizons; no decay in ~17 years. Fades past ~15–20 days; understates regime breaks.	Yes — as risk bands and position sizing, not as trades	[12]
52-week-high proximity	MIXED	~0.45%/mo in-sample 1963–2001; fails in Japan; weakened post-publication. Note the direction: new highs historically predicted *continuation* — "new highs are bearish" is backwards.	Context signal only	[15]
High-volume return premium	MIXED	Unusual volume carries information (~0.5%/mo strongest deciles, 1963–96); concentrated in small caps with the widest spreads.	As information, not as a strategy	[16]
Short-term reversal (buy recent losers)	MIXED	Huge gross returns (~2.5%/mo, 1934–87) that are really the market makers' wage for liquidity provision. Naive versions die after costs; cost-aware large-cap versions earn 30–50bp/week — with institutional execution.	No — you are the counterparty, not the market maker	[17,18,19]
Classic chart patterns (H&S, double bottoms)	MIXED	Kernel-regression tests find some patterns shift the return distribution — genuine but faint informational content; no profitable strategy demonstrated. Frequently miscited as proof charting "works."	No demonstrated net edge	[20]
Support/resistance at round numbers	MIXED	Best mechanism in TA: real order clustering at round numbers produces bounces ~4.6pp more often than arbitrary levels — but the horizon is intraday and no cost-surviving strategy is documented.	Explains "levels"; not tradable	[21]
Index moving-average / range-break rules	WEAK	Beat the Dow in-sample 1897–1986; the best of 7,846 rules showed nothing out-of-sample 1987–96 after data-snooping correction; nothing net of costs through 2011 in any ex-ante-selectable period.	No	[22,23,25]

Technique	Verdict	What the careful evidence shows	Retail capture?	Refs
Faber 10-month MA trend filter	MIXED	A drawdown-reduction tool, not a return enhancer: historically cut max drawdowns roughly in half, lagged buy-and-hold post-2008, loses to whipsaws in V-recoveries. Honest frame: crash insurance paid for in return.	Yes, as insurance — if the premium is understood	[27,28]
RSI and standard oscillators	WEAK	No significant net profits at standard settings in developed markets; positive findings come from parameter tuning that invites snooping.	No	[29]
Candlestick patterns	WEAK	28 patterns on DJIA stocks 1992–2002: no value before costs; Japan replication negative. Contested-negative, not unanimous: some three-day patterns show profits under alternative exit rules — small, specification-sensitive, unlikely to survive retail costs.	No	[30,31,32]
Fibonacci retracements / Elliott Wave	FOLKLORE	Bounce rates at Fibonacci zones statistically indistinguishable from arbitrary zones; Elliott counts revisable after the fact are largely unfalsifiable. Round-number clustering explains apparent "levels."	—	[33,21]
"Gaps always fill"	FOLKLORE	The main peer-reviewed gap study finds momentum in the gap's direction, not reversal. The "70–90% fill" statistic has no traceable source; eventual fills are a base-rate artifact.	—	[34]
Daily-direction prediction (ML / signals)	WEAK	Sound out-of-sample tests: 50–55% vs an always-up baseline of ~53–55%. A "58%-accurate" signal can literally be worse than always guessing up. Pre-2010 edges decayed.	Treat every daily prediction as entertainment	[24,69]
Index timing via valuation/macro dials	FOLKLORE	Fails against the naive historical mean out-of-sample; a slow valuation signal exists at decade horizons but cannot time entries.	—	[35]
Volatility-managed portfolios	MIXED	Headline Sharpe gains failed direct out-of-sample tests across 103 strategies; the risk-stabilization benefit (steadier ride, tamer left tail) is real because volatility is forecastable.	As risk control, not return booster	[36]
Factor tilts via low-cost funds	MIXED	Most factor themes replicate globally *and* decay 30–60% post-publication; most published factors fail the t>3.0 hurdle. Expect a fraction of advertised returns.	Yes — through cheap funds	[10,8,9]
Overnight-vs-intraday return split	MIXED	Statistically robust (the equity premium historically accrued close-to-open) and a failed product: the NightShares ETFs underperformed by ~29pp and liquidated within 13 months.	Object lesson: significance ≠ exploitability	[37,38]
Volume/attention spikes as buy signals	WEAK	Attention is a hazard: the stocks retail crowds bought most intensely earned −4.7% abnormal returns over the following 20 days; search spikes pop then revert.	Inverted, if anything — the app flags these as risk	[50,51,52]

4.1 Notes on the rows that matter most

Momentum is the honest poster child. The strongest TA-adjacent effect in the literature also carries the ugliest tail: two-to-three-month drawdowns of −73% to −91% (gross), short-side-driven, clustered in rebounds after high-volatility bear markets[14]. Any presentation of momentum without its crash profile is marketing. The app pairs them permanently.

The moving-average lesson generalizes. The Brock–Lakonishok–LeBaron rules are the canonical demonstration of why in-sample success means little: honest re-testing across the full universe of 7,846 rule variants erased the edge the moment the sample ended[23]. This is why every base rate in the app carries a trial-count correction (Part 8) — with five years of daily data, trying ~45 configurations essentially guarantees an impressive, worthless in-sample winner.

Candlesticks illustrate calibrated honesty. The cross-examination pass in this project's research flagged that "cleanest folklore verdict" overstated the record: one credible study finds some three-day patterns profitable under different exit rules[32]. The accurate label is *contested-negative* — and that is exactly how the app's Academy teaches it, because modeling honest gradation is part of the curriculum.

"Gaps always fill" is the archetypal folklore trap: intuitive, everywhere on YouTube, and pointing the wrong way — the one serious study finds continuation, not fill[34]. The Academy uses it as the worked example for "how to check a claim against a base rate."

PART 5

The catalyst playbook

Catalysts are where preparation beats prediction. The app cannot know which way an event breaks — but it can know the event is scheduled, how large the move typically is, and which patterns around it are real versus invented.

CATALYST BEHAVIOR — WHAT IS ESTIMABLE, AND THE TRAP TO AVOID

Catalyst	Typical behavior (with base rates where known)	The trap	Refs
Earnings announcement	Mega-caps typically move 3–6% absolute on earnings day, small caps more. Reactions are asymmetric: negative surprises punished far harder than positive ones are rewarded — worst for high-expectation growth names (the “torpedo effect”).	Betting the direction of the reaction; it is not predictable from the setup	[39,40]
Forward guidance	Guidance frequently dominates the EPS print: mean reaction ≈ −10% to a cut vs only ≈ +2% to a raise. A company can beat and still fall hard.	Reading the headline beat/miss as the story	[41]
Options-implied move	The ATM straddle is a roughly unbiased estimate of move *size*. Pricing is name- and regime-dependent: realized falls short of implied 40–75% of the time depending on the name — above half (modestly rich) for many mega-caps, the reverse for several high-volatility names.	Blindly buying *or* selling options into earnings — neither survives retail spreads reliably	[39,42]
Post-earnings drift (PEAD)	Historically ~4% top-minus-bottom spread over 60 days; essentially zero in large caps since ~2006 as prices adjust on announcement day. Survives mainly in microcaps where costs consume it. (2025 papers contest the obituary — see Part 10.)	Expecting exploitable drift in liquid names	[43]
FOMC meetings	The famous pre-FOMC drift (+49bp in the 24h before announcements, ~80% of the annual equity premium 1994–2011) vanished after ~2015, post-publication.	Trading a decayed anomaly still circulating in blog posts	[44,45]
Macro prints (CPI, NFP, GDP)	Intraday volatility spikes ~3–6× normal in the minutes around release; full-day vol only modestly elevated. Markets react to revisions, composition, and policy implications — not the headline alone.	Predicting direction from the headline surprise	[46]
Analyst rating changes	≈ +3.0% (upgrades) and −4.7% (downgrades) over three days at announcement; the jump happens within minutes.	Chasing the news after the move — nets retail little or nothing	[47]
M&A announcement	Target premiums average ~20–30%, captured almost entirely in the instantaneous gap; the residual spread is professional deal-risk compensation.	“Merger arbitrage” as a beginner opportunity	[48]
FDA / biotech binaries	Single-asset small caps move 40%+ either way on PDUFA outcomes; large pharma barely moves on the same news. A genuine coin flip to outsiders.	Confusing high volatility with high opportunity	[49]

Catalyst	Typical behavior (with base rates where known)	The trap	Refs
Attention spikes / trending lists	The most intensely crowd-bought stocks earned −4.7% abnormal returns over the following 20 days; search spikes pop ~2 weeks then revert within the year.	Treating "most bought" lists as discovery — the app deliberately has none	[50,51,52]
News sentiment / LLM headline scores	Statistically real but tiny and concentrated in small/illiquid names and the short leg; excludes costs; decaying as adoption spreads — a live decay experiment.	Using sentiment as a trade signal rather than context	[53,54]
No news at all	Half to two-thirds of large daily moves have no cleanly identifiable cause: even modern text methods tie only ~33% of return variance to news days vs 16% on no-news days.	Inventing a narrative anyway — the app must say "no clear catalyst found; likely noise"	[55,56]

REALITY CHECK

The single most protective row is the last one. Financial media produces an explanation for every move by 9:00 the next morning; the evidence says most of those explanations are invented after the fact. An app that is willing to display *"no clear reason — most moves this size have none"* is teaching its user something most professionals never say out loud.

PART 6

The retail reality

These are not anecdotes or survivor stories — several are full-population censuses, which makes “most day traders lose” one of the best-evidenced facts in all of finance. The app is designed to keep its user out of these statistics.

Underperformance of the most active retail households	Taiwan day traders persistently profitable, net of fees	Brazilians who day-traded 300+ sessions and lost money	Return after a stock hits the “most bought” list
−6.5 pp/yr	**<1%**	**97%**	**−4.7%**
Barber & Odean, 66k accounts [26]	full-population census [57]	no learning-by-experience found [58]	next 20 days, abnormal [52]

6.1 The census numbers

Three studies anchor everything. Barber & Odean’s analysis of 66,000 US brokerage households found the quintile that traded most underperformed by about 6.5 percentage points per year — the drag almost exactly equal to their trading costs[26]. The Taiwan census — every day trader in an entire national market over 15 years — found fewer than 1% persistently profitable net of fees; the winners showed *pre-existing* skill, not learned skill[57]. The Brazil census of futures day traders found 97% of those who persisted 300+ sessions lost money, with no evidence that experience improved outcomes[58] — directly contradicting the education industry’s core promise that day-trading profitability is a learnable craft.

6.2 The behavioral taxes

- **Disposition effect:** investors sell winners and keep losers; the winners they sold went on to beat the losers they kept by 3.4pp over the following year[61].
- **Lottery preference:** the tilt toward cheap, high-skew “jackpot” stocks costs roughly 2–3pp per year[63].
- **Overconfidence:** men trade 45% more than women and pay roughly 0.9pp/yr more in trading drag for it (single men vs single women: 67% more trading, ∼1.4pp)[62].
- **Attention herding:** buying what is trending is reliably value-destroying (−4.7% over 20 days for the most crowd-bought names[52]).

None of these are exotic; all four will knock on this user’s door in month one. The Academy teaches each one *through the friction surfaces that detect it* — the cost mirror, the trade-frequency mirror, and the cooling-off prompts on the paper-trade ledger — a named exception to the Academy’s pull-only rule (§9.5), rate-limited and dismissible. That pairing is a core product mechanic, not a nice-to-have.

6.3 Microstructure a small investor can actually use

Three pieces of execution knowledge are free money relative to ignorance: (1) spreads are widest in the first ∼30 minutes — over 20bp on S&P 500 names at the open versus under 5bp late in the day, per industry measurement — so avoid market orders at the US open; (2) prefer limit orders as a default; (3) the closing auction concentrates 10%+ of daily volume and is the deepest window of the day[66]. This is the rare category where knowledge converts directly into saved basis points.

6.4 The regulatory ground shifted in 2025–26

Two recent changes matter to the design. First, FINRA eliminated the pattern-day-trader $25,000 rule effective June 4, 2026, replacing it with broker-set intraday margin[59] — most web content still describes the old rule as law; the app must not repeat it, and more importantly, **the last external brake on beginner overtrading is gone, so the app supplies its own friction** (trade-frequency mirror, cooling-off prompts, cost projections). Second, the SEC withdrew the Order Competition Rule and Reg Best Execution in June 2025[60]: payment for order flow remains legal and disclosure-based, and "price improvement" claims should be read against the midpoint, skeptically.

> **ASSUMPTION & POSTURE — THE USER**
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> The user is a US resident (Long Island, New York — Eastern Time, the market's own clock). Nothing restricts their access to US equities; any retail brokerage would take the account tomorrow. The design nonetheless **defaults to paper-trading and watchlist mode and ships no brokerage onboarding in v1 — a purely pedagogical stance**: the censuses above price the tuition of learning with real dollars, the last external brake on beginner overtrading (the PDT rule) was removed in June 2026, and every honesty mechanism in this product — the graded track record, the cost mirror, the calibration score — works identically on paper. Live-brokerage integration is therefore a legitimate *later* extension, gated on the paper record demonstrating readiness — an execution decision, not a legal question.

6.5 Free data has a survivorship problem

One quieter trap belongs in this part because it corrupts self-run research: free price datasets typically include only currently listed companies. Missing performance-delisting returns average about −30%[65], so backtests on survivor-only universes flatter every strategy — by 1.5+ pp/year in small caps. The app warns whenever a backtest universe excludes delisted names, and the Blueprint's data model carries delisting status for exactly this reason.

PART 7

What ML and LLMs genuinely add

The after-close pipeline uses machine intelligence heavily — but in exactly one role: a grounded reader, summarizer, and explainer of things that already happened. Here is the evidence for why the line is drawn there.

7.1 The ceiling on ML prediction, measured

The best-in-class academic result — Gu, Kelly & Xiu's neural networks over decades of US data — achieves an out-of-sample R^2 of about **0.4% per month per stock**[67]. Read that carefully: 99.6% of monthly variation remains unpredictable, and the economic value of the predictable sliver materializes only in institution-scale, diversified long-short portfolios heavy in microcaps. Follow-up work imposing economic restrictions (exclude microcaps and distressed names, apply realistic costs) cuts the risk-adjusted gains by more than half and erodes the rest[68] — the retail-tradable residual is approximately zero. The celebrated deep-learning daily-direction papers fare worse: the documented pre-2010 LSTM edge decayed to roughly zero after costs post-2010[69], and high-accuracy claims typically dissolve into leakage, lookahead bias, or denoising artifacts on inspection[24].

7.2 Where LLMs are strong — and where they fail in ways that matter here

Capability	Evidence & consequence for this product
Summarizing filings & calls STRONG	Model-written summaries of disclosures explain market reactions *better* than the original documents[71]. This is the pipeline's core LLM job.
Sentiment classification CAREFUL	Competitive with specialized models when prompts are engineered; naive prompting can score below logistic regression. Anonymizing entity names *improved* live performance by removing memorized associations[70].
Numeric work FAILS	Models collapse from ~95% accuracy on simple financial lookups to near 0% on multivariate calculations[73]. Consequence: **the pipeline computes every number; the LLM narrates them verbatim** and a deterministic gate verifies each one appears in the sources.
Verbalized probabilities SUPPRESSED	LLM-stated confidences are systematically overconfident across model families[74]. The app never displays a model's own probability as a market probability; all probabilities on screen are frequency counts from historical data.
Backtesting LLM signals TRAP	Training-corpus memorization is a leakage channel unique to LLMs: any backtest on pre-cutoff data is untrustworthy[70]. Post-cutoff evaluation is the minimum standard; the headline-sentiment edge is itself decaying as adoption spreads[54].
Autonomous trading agents RULED OUT	Published agentic-trading frameworks show single-regime evaluations, leakage, and pathological overtrading; no credible durable alpha. Legitimate for research automation only.

THE ROLE, STATED ONCE

The LLM in this product is a **grounded explainer, disclosure summarizer, and screener-narrator —** with retrieval grounding, forced per-claim citations, deterministic numeric verification, and suppressed verbalized probabilities. It is never a price or direction oracle. The Blueprint implements this as a three-stage pipeline with a hard publish gate.

PART 8

From evidence to design

Everything above compresses into two lists: product commitments (what the app does and refuses to do) and statistical guardrails (rules the software enforces mechanically, so honesty doesn't depend on daily discipline).

8.1 Product commitments

- **Never ship a directional forecast.** The only quantitative prediction is a short-horizon volatility band (≤ ∼1 month), drawn from fat-tailed distributions and labeled as a conditional estimate that can understate sudden stress.
- **Every base rate carries its sample size and a Wilson confidence interval.** "62 of 110 cases" with an interval — never a bare "56%".
- **Every published pattern carries a decay stamp:** publication date plus the expected post-publication haircut, so "proven strategies" arrive pre-shrunk.
- **Movers never appear without a catalyst check.** If no catalyst is found, the row says so — "no clear reason; most moves this size have none" — instead of inventing one.
- **No trending/most-bought discovery surfaces, ever.** That feature causally generates losses[52]. High-skew, low-price tickers get lottery-risk flags instead.
- **The cost mirror is a first-class widget:** projected annual drag = effective spread × the user's own turnover, plus a trade-frequency mirror and cooling-off prompts — the app's replacement for the abolished PDT brake.
- **Position-size suggestions (paper portfolio only) are capped at half-Kelly**, computed from the confidence-interval *lower bound* of the win rate together with the stored payoff distribution (forward p10/median/p90) — a teaching device for sizing arithmetic on paper, never a conviction dial, and it inherits every cost caveat on the card.
- **The app grades itself in public.** Every flagged setup is logged append-only and scored after resolution; the track-record page shows hits, misses, and calibration — misses first-class.
- **Folklore is labeled folklore** — Fibonacci, Elliott Wave, gap-fill lore, and indicator-based market timing are taught with their evidence grade, not silently omitted.
- **Every real effect names who can capture it** (institutions at scale vs individual via funds) and at what horizon — the "real but not yours" frame from Part 2.
- **Paper-first by default** — a pedagogical stance, not a legal one (§6.4); no brokerage integration in v1, revisited only after the paper track record earns it.

8.2 Statistical guardrails (enforced in code)

Rule	Rationale
t > 3.0 before any new pattern is labeled better than noise	Multiple-testing hurdle for a mined field[9].

Rule	Rationale
Trial counting on every user experiment (Deflated Sharpe, PBO)	~45 tried configurations on 5 years of daily data guarantee an impressive in-sample mirage[72].
Sample-size display rules: N ≥ 100 → % + CI; 30–99 → "roughly 6 in 10" + wide-interval note; N < 30 → "insufficient history — treat as anecdote," percentage suppressed	60/100 wins spans [50%, 69%] at 95% confidence — visibly near a coin flip; false precision is the enemy.
Direction benchmarks use the always-up baseline (~53–55%), never 50%; anything above ~55% OOS is flagged as presumptive overfitting	US equities drift upward; a "58% accurate" signal can be worse than "always up"[24].
Decay haircuts auto-applied to published anomalies (−26% OOS / −58% post-publication, US)	McLean–Pontiff averages[8]; haircut noted as US-calibrated.
Survivorship warnings when a backtest universe lacks delisted names	Missing delisting returns average ~−30%[65].
Retail cost assumptions (20–57bp/trade) in every net-of-cost figure; strategies above ~50% monthly turnover presumptively die	Institutional cost assumptions are the standard silent lie in backtests[3].
Volatility forecasts only at ≤ 15–20 trading days, with regime-break caveat attached	That is where the evidence ends[12].
Regime-sliced statistics get wider bands or are blocked below minimum N	Conditioning multiplies implicit tests while shrinking samples.
The app's own flags are Brier-scored; the forecast journal anchors user predictions on displayed base rates	Calibration feedback is the best-evidenced debiasing intervention available.

PART 9

The product: Desk & Academy

One product, two rooms. The **Desk** is a calm, one-screen command center that runs the daily ritual over the US trading day. The **Academy** is a prose-first learning hub. They are visually and structurally separate, connected only through named doorways — and every element the evidence chapters demanded (base rates, decay stamps, honest misses) has a specific home here.

9.1 One product, two rooms

The hard requirement — data experience and learning experience cleanly separated, neither crowding the other — is solved with **two distinct visual grammars under one shell**, so the user knows which mode they are in without reading a label[75,90]. The Desk is dense and tabular: compact 13–14px type, cool neutral background, tabular lining numerals, optional dark mode. The Academy is editorial: 16–18px prose at a ~65-character measure, generous line height, warmer white, light-only (long-form comprehension favors positive polarity[77]). Two content rules keep the wall solid: *the Desk hosts doorways, not classrooms* — chips, glossary popovers, and one worked-example drawer, but never full lesson bodies; and *the Academy contains no live prices*, so the monitoring mindset and the learning mindset never compete for the same screen.

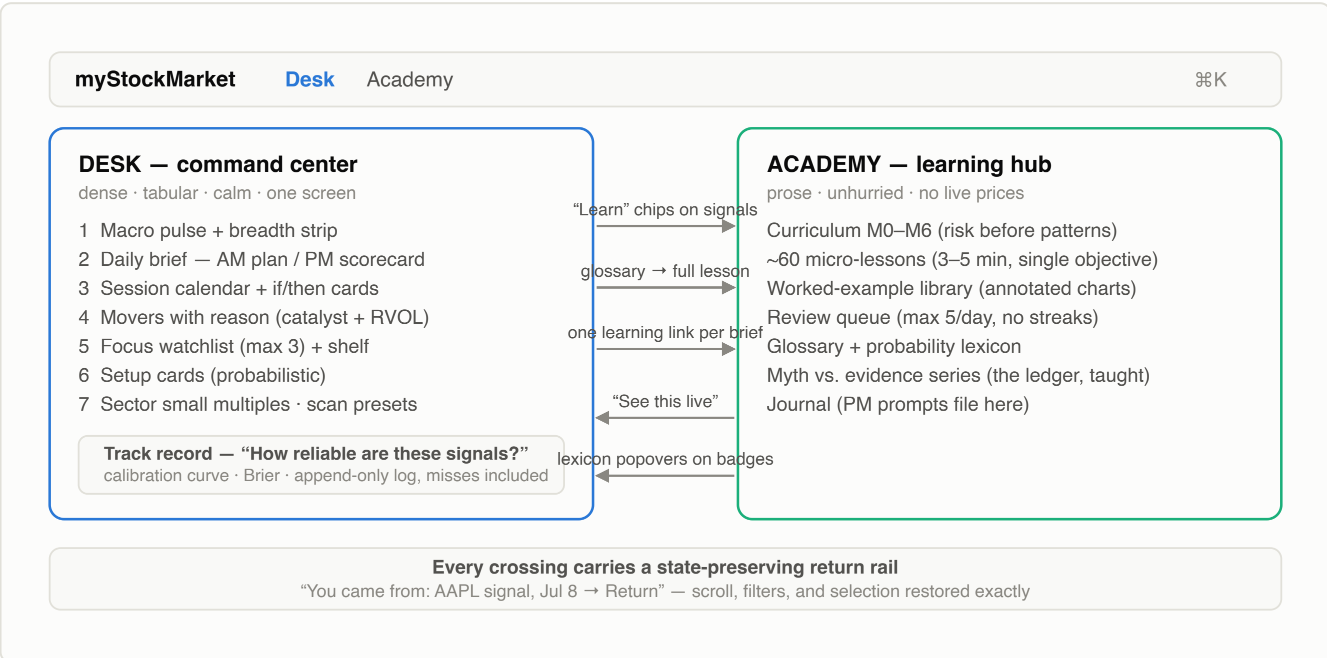

Figure 9.1 — Information architecture. Two zones with distinct visual grammars; guaranteed clean return. Seven crossings share the return-rail contract: the five doorway arrows shown, plus the worked-example drawer (disclosure level 2 of every signal, §9.3) and the PM journal prompt, which files into the Academy journal. The behavior-mirror prompts of §6.2 (cooling-off, frequency mirror) are the one sanctioned *push* channel — rate-limited, dismissible, each carrying a single lesson link [79,80,94].

9.2 The Desk and the daily ritual

Behind the ritual sits the after-close engine. Each trading evening — hours after the 4:00pm ET close, entirely in the cloud — a pipeline ingests the day's US closing data, the earnings calendar and filings, macro releases and ticker-tagged news; scans the full US market for pattern states; computes base rates with their intervals; and writes — then deterministically verifies — the briefing. By roughly 8:40pm Eastern the Desk is already set for tomorrow, whether it gets read that night or over pre-open coffee. (The engine's full design is Blueprint Parts 2 and 5; everything it publishes obeys the guardrails of Part 8.)

The Desk fits one screen without scrolling[75] at a reference viewport of ≥1366×768; below that, modules 6–8 collapse to summary rows and the page scrolls *in ritual order* — the order, not the single screen, is the invariant (on a phone, the Desk renders as one ritual-ordered column). That fixed top-to-bottom order mirrors the documented professional pre-market sequence — macro first, calendar, movers, then the plan — so the layout itself teaches the routine[90,96]. Positions and P&L are deliberately *not* the hero: institutions never start the day with their book, because it anchors emotion before context[96].

The timing geometry is home-field: the user lives on market time. The regular session runs 9:30am–4:00pm ET with after-hours until 8:00pm ET; the verified briefing publishes by about 8:40pm the same evening (an hour earlier on the winter clock — the pipeline is UTC-fixed while local time shifts with DST), hours after the close and more than twelve before the next open. The ritual is therefore a **review-and-prepare** session — built strictly from closed-session data, never a live feed — bookended by two fixed-format briefs[91]: an AM plan of three to five word-capped items with labeled slots (*What happened / Why it matters / By the numbers / Yes, but*) covering the completed US session, and a PM scorecard that grades the prior brief's flags against what actually happened, misses included. That evening loop is the strongest anti-hype widget in the product: the user watches base-rate intuition form against their own tool's record. The brief's lead card — *Today's focus* — together with any strong-tier setup cards is the explicit answer to "what deserves a deeper look"; and "no clear edge today" renders as a first-class, undecorated state, never an apology.

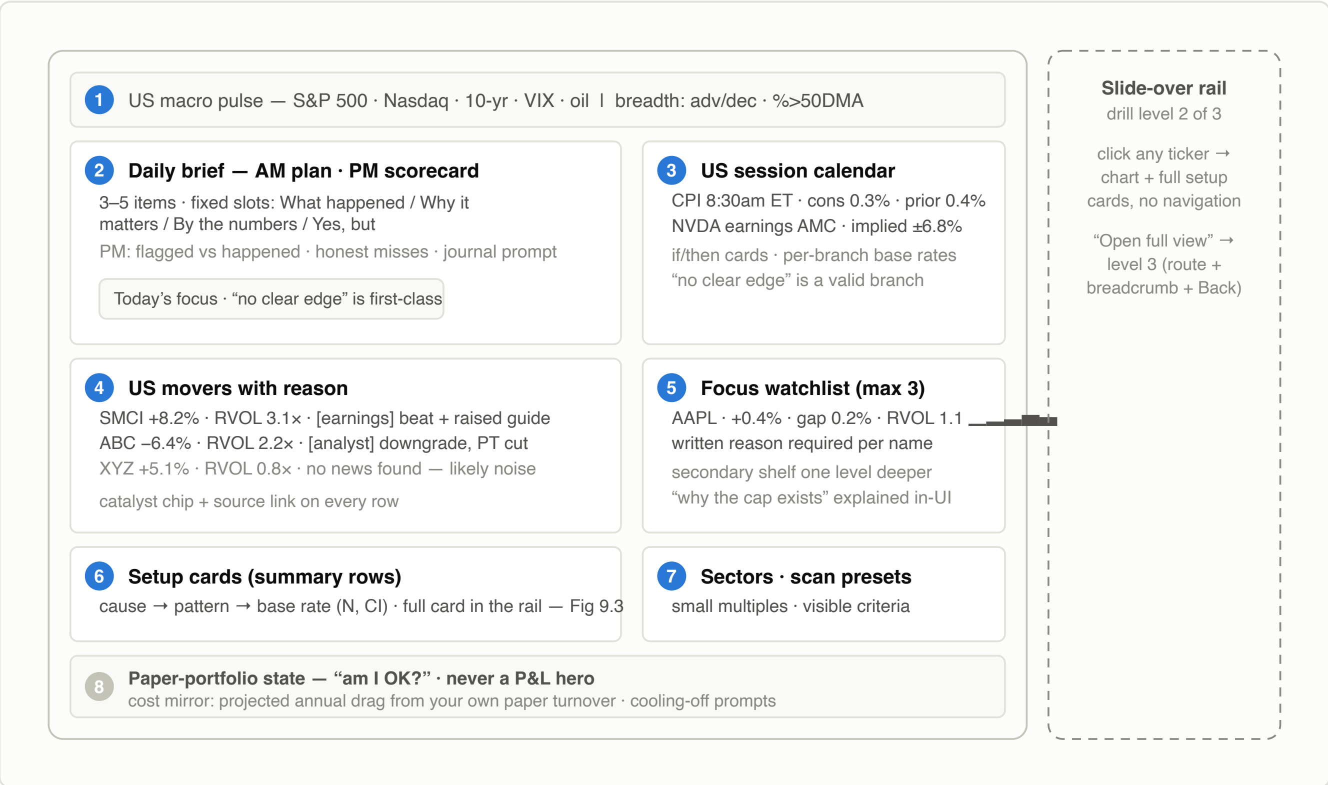

Figure 9.2 — The Desk, one screen. Numbered in ritual order (macro → brief → calendar → movers → watchlist → setups). Setup cards render as summary rows here; the full template (Figure 9.3) lives in the slide-over rail — drill depth 2. A full ticker page with breadcrumb is depth 3, the maximum [77,98].

WIDGET CATALOG — EVERY MODULE EARNS ITS PLACE OR DOESN'T SHIP

Widget	Priority	Why it earns its place	Explain-why hook
US macro pulse + breadth strip	MUST	Every documented pro routine starts macro-first; breadth answers "broad or narrow?" — invisible in price-only views [90,96].	Every figure is a glossary term; popover explains disagreement cases.
Daily brief pair (AM/PM)	MUST	Fixed-count, fixed-order briefs readable in under 3 minutes are the proven ritual format; labeled slots separate fact from interpretation structurally [91].	The slot labels are the lesson; exactly one Academy link per brief.
Movers with reason (US session)	MUST	The highest-leverage terminal pattern for a beginner: why + relative volume on the same row; catalyst-less moves labeled "likely noise" [92].	Catalyst chip carries durability + continuation base rate; opens catalyst explainer.
US session calendar + if/then cards	MUST	Converts surprise volatility into anticipated events (earnings bmo/amc, CPI, jobs report, FOMC); scenario framing beats point forecasts, and "no edge" is a rendered branch [84].	Consensus/prior columns teach beat/miss; per-branch frequencies (suppressed honestly when branch N < 30).
Focus watchlist (max 3 + shelf)	MUST	Pros cap attention deliberately; unbounded lists produce scattered emotional decisions [96]. RVOL is a base rate disguised as a column [99].	The cap itself is explained; headers are glossary terms.
Probabilistic setup card	MUST	The signature unit — see §9.3 and Figure 9.3.	Two-level disclosure: chip → worked-example drawer → Academy.
Track-record page	MUST	Grading itself in public — misses first-class — is the app's structural advantage over hype [94,80]. Ships (in minimal form) with the first setup cards, never after them.	Companion lesson uses the live page as its worked example.
Sector small multiples	SHOULD	Identically scaled mini-charts make "broad or concentrated?" preattentive [76]; desaturated — the saturation-wall look is rejected as emotional wallpaper.	Tiles link to sector-rotation explainers.
Quantile dotplot (drill-down)	SHOULD	Each dot = one past occurrence's forward return; the losing tail is visibly present. Measurably reduces lay estimation error vs point numbers [85].	One-time "how to read a dotplot" signpost, then silent.
Named scan presets, visible criteria	SHOULD	The visible recipe doubles as curriculum; hidden criteria turn a preset into an oracle [78].	Each filter line links to the lesson explaining it.
Curated top news (not a feed)	LATER	Curation works; a firehose is cognitively hostile and mostly duplicated by movers-with-reason.	Neutral declarative headlines; catalyst chips consistent with movers.
Relative-strength comparison chart	LATER	Good on the full ticker page; kept off the Desk to protect the widget budget [98].	"Relative strength" glossary chip.
⌘K command palette	LATER	Zero-chrome accelerator; zone badges on results reinforce the two-room model.	—

9.3 The setup card — the signature unit

Wherever a signal fires, it renders in one fixed template. On the Desk itself a card appears as a two-line summary row; the full template below is what the slide-over rail shows. The template is the XAI research distilled[79,80]: an observable cause (never "our model likes it"), a named pattern that links to its lesson, a base rate stated as a **natural frequency with an explicit reference class**[81,82], the sample size and interval, an explicit comparison to the unconditional baseline, a typical range instead of a target, a mandatory "what would weaken this" list (explanations inflate trust even when wrong — the falsifiers push back[80]), and a scope line. Verbal labels come from a fixed lexicon[83] — "moderate tendency" always means the same numeric band, documented once in the Academy.

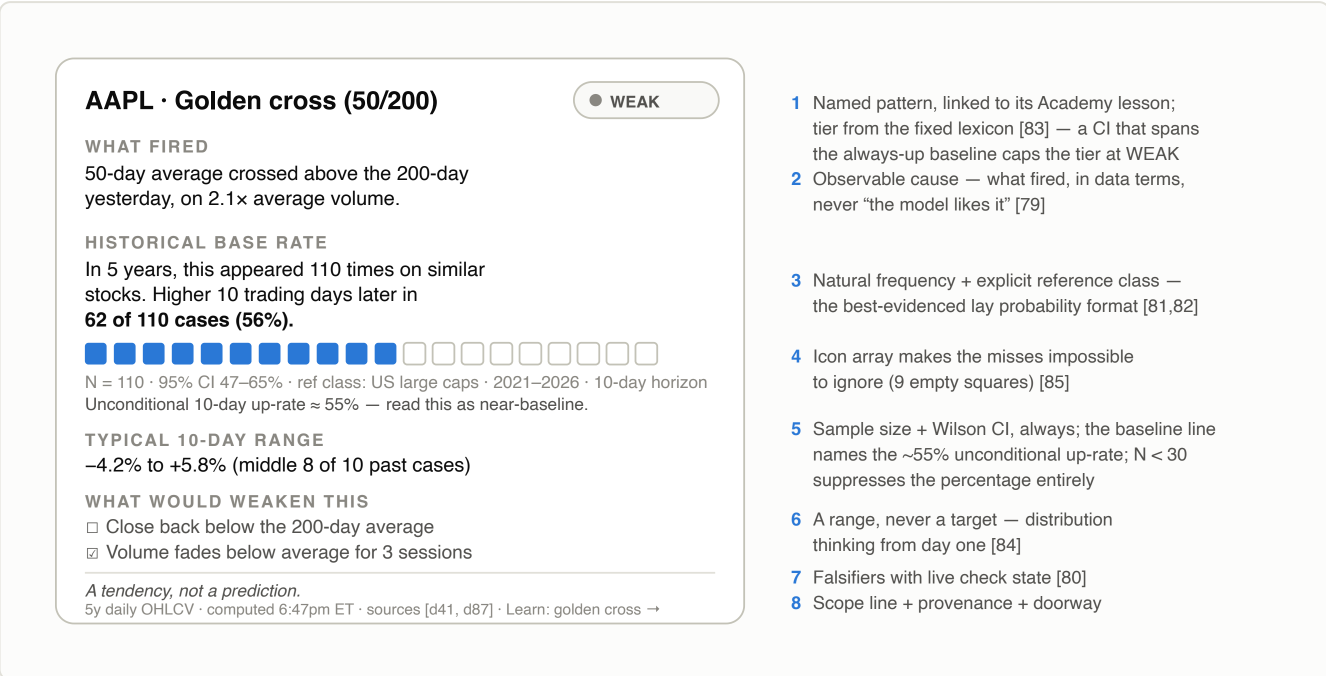

Figure 9.3 — Anatomy of the probabilistic setup card. The fixed template renders identically for every signal; only the content varies. Note the example's own honesty: 56% against a ~55% unconditional up-rate, with a CI touching the coin-flip line, earns only a WEAK tier — the flagship illustration is deliberately a near-non-signal. Clicking "Why this fired" opens a three-step worked-example drawer with the relevant candles annotated on the chart itself — disclosure level 2 of a hard maximum of 2 overlays [78,86].

9.4 The honest-uncertainty system

- **The canonical sentence.** Every base rate reads: *"In the past 5 years, this pattern appeared N times on stocks like this. Price was higher H days later in W of N cases (P%)."* Natural frequencies with a visible reference class — because 30–80% of laypeople misread bare percentages by misattributing what they refer to[81,82].
- **Sample-aware precision, enforced by the display engine.** N ≥ 100 → percentage with interval; 30–99 → "roughly 6 in 10" with a wide-interval note; N < 30 → "insufficient history — treat as anecdote," percentage suppressed. Never "68.4%" from 12 occurrences.
- **Words from a fixed lexicon.** "Weak tendency" = 50–58%, "moderate" = 58–70%, "strong" = >70% — always rendered word *plus* frequency, because interpretations of a bare "likely" span 40+ percentage points across readers[83]. One override outranks the bands: **a confidence interval that includes the ~53–55% always-up baseline caps the tier at "weak"** — no-edge never wears a stronger badge, whatever the point estimate says.

- **Distributions, never points.** Quantile dotplots of historical forward returns in drill-down[85]; fan charts with frequency-labeled bands ("8 in 10 past paths stayed inside") wherever anything forward-looking is drawn[84]. No single projected line exists anywhere in the product.
- **Confidence only where it changes behavior.** Tier badges appear on actionable setup cards only, each tier mapped to a stated, deliberately observational action — "strong → worth a closer look; moderate → note it and check the weakeners; weak → watch only." No tier ever maps to an acquisitional verb, and no confidence chrome appears on quotes or static facts — ubiquitous scores train score-chasing[79].
- **The system grades itself in public.** Calibration curve with per-bucket counts, rolling Brier score with a plain-language anchor ("0.25 = coin flip"), and a complete append-only resolved log — every hit *and* miss, filterable by pattern, with a banner when a signal family runs below its long-run rate[94].
- **Mechanical voice.** "The pattern has historically been followed by…" — never "I think this stock will…". Anthropomorphic voice inflates trust; disclaimers sit at the decision point, not the page footer[77,97].

9.5 The Academy

The Academy is a curriculum, not a content dump: seven modules of ~60 single-objective micro-lessons (3–5 minutes, two or three retrieval questions, a quiet checkmark — chunking without gamification). The ordering is deliberate: **risk before patterns**, with pattern lessons soft-gated on the risk module — teaching pattern-spotting first manufactures overconfident beginners, the exact failure mode this product exists to prevent.

CURRICULUM MAP

Module	Contents
M0 · Orientation	How this app explains itself: the probability lexicon, reading base-rate sentences and icon arrays, what the track-record page means (calibration, Brier as "coin flip = 0.25").
M1 · US market mechanics	NYSE and Nasdaq; tickers; the US session structure (pre-market 4:00–9:30am, regular 9:30am–4:00pm, after-hours 4:00–8:00pm ET) and what trades when; order types; bid–ask spreads; what every number in the macro strip is.
M2 · Costs & account math	Spreads, slippage, taxes, why frequent trading compounds costs — the arithmetic that precedes any strategy.
M3 · Risk before patterns	Position sizing, stops, expectancy, drawdown math, why base rates beat anecdotes. Gates the pattern modules.
M4 · Reading charts	Candles, support/resistance, volume and RVOL, gaps, sparklines and small multiples.
M5 · Indicators & signals	Exactly and only what the Desk shows: each pattern's evidence grade from Part 4, how each base rate is computed, every scan preset's recipe. The "myth vs. evidence" series lives here.
M6 · Psychology & biases	Base-rate neglect, chasing, disposition, overtrading, why the focus list is capped at 3; journaling practice — the PM prompts file here.

Four mechanisms make it learning-in-context rather than a separate course. (1) **Worked examples**: every signal's "why this fired" drawer is a fixed three-step solved example — novices learn far more from worked examples than raw data[86] — with annotations drawn *on the live chart*, synced to the steps, because separating explanation from chart splits attention[86]. (2) **Pull, not push**: no tours, no onboarding wizard; the one push moment is a one-time first-encounter signpost per concept[78] — plus the single named exception of §6.2's behavior-mirror prompts (cooling-off, frequency mirror), which are rate-limited, dismissible, and carry at most one lesson link. (3) **Spaced review**: at most five daily retrieval prompts, Leitner-scheduled, seeded only from concepts the user actually encountered on the Desk that week — spacing is one of the largest replicated effects in learning science (d ≈ 0.6)[88] — skippable, guilt-free, no streaks. (4) **Adaptive fading**: mastered concepts collapse to terse pro-mode and worked examples flip into completion prompts — permanent hand-holding measurably harms intermediates (the expertise-reversal effect[87]), and a tool still explaining candlesticks in month six gets abandoned. Inside the Academy, navigation stays shallow: curriculum map → lesson (breadcrumbed), glossary entries open as popovers within a lesson, and the return rail renders whenever arrival was from the Desk.

9.6 Navigation

- **Drill depth is capped at 3:** glance (Desk) → slide-over rail (~420px, in place, no navigation) → full page (own route, breadcrumb, fixed "Back to Desk")[77].
- **Return is guaranteed and stateful:** scroll, filters, and selection restored exactly; a beginner who gets lost twice stops drilling at all[78]. Cross-zone jumps carry the persistent return rail (Figure 9.1). No modal stacks.
- **High-scent labels only:** every link states the question it answers ("Why is NVDA moving?"), never "Details"[78].
- **Overlay budget:** chips, tooltips, popovers, and the one slide-over drawer — anything longer navigates to the Academy. Two disclosure levels, never three[78].
- **Constrained customization:** fixed zone skeleton with show/hide and reorder within zones; no freeform drag-anywhere — it destroys the hierarchy the ritual depends on[77].

9.7 Visual language

> AMENDED 2026-07-11
>
> By deliberate user directive, the austere clauses of this section (no shadows, no gradients, neutral-only chips, motion minimalism) are relaxed; the operative visual law is now **UI-REDESIGN-PLAN.md** ("Morning Broadsheet": gradients, soft elevation, rounded glass cards, colored chips, and general UI motion are allowed, all tokenized and grepped). The clauses below are annotated in place. Unchanged and non-negotiable: probability visuals are motionless; gain/loss is colorblind-safe and never color-alone; amber stays reserved; position-and-length over angle-and-area; dark mode stays Desk-only. §9.8's anti-pattern table stands in full — every row of it was honesty, not taste.

- **Data-ink maximalism — amended 2026-07-11:** soft tokenized elevation and glass surfaces are now allowed; the heuristic that survives is the test itself — if an element can disappear without losing data or structure, it does[76,75]. Decoration must never impersonate data.
- **One reserved attention color** (amber) for active alerts — at most 2–3 visible before they aggregate into a count. Color communicates importance only when scarce[75]. *Amended 2026-07-11:* tier badges are now colored soft-wash chips with their own hues (never the alert amber) and always carry their word — the amber budget still cannot be spent by tiers, so ten firing cards still cannot turn the Desk amber.
- **Semantic-only, colorblind-safe gain/loss:** *amended 2026-07-11* — the pair is now blue #2563EB / orange #EA580C (with darkened AA text steps; previously Wong #0072B2/#D55E00[93]), applied only to deltas and chart lines, always redundantly encoded with sign and direction; banned from buttons, links, and decoration — interactive elements use a distinct link color so a delta can never be mistaken for a button. The rule is unchanged; only the hues moved.

- **Tabular lining numerals** (a UI sans with `tnum` support — Inter or IBM Plex Sans class), right-aligned numeric columns, consistent decimals — aligned digits scan instantly. Reference tokens for the coding phase: Desk background near-white cool neutral, Academy warmer white, one hairline divider tone, a 4–5 step type scale per zone; exact values are the first task of the build's design pass, constrained by this section.
- **Calm technology — amended 2026-07-11:** general UI motion (page transitions, hover states, one-time chart draw-ins, micro-interactions) is now allowed; calm is preserved as *no blinking, no ticker tape, no toasts, no badges, no loops, no autoplay, no urgency* — and in v1 the data is as-of the last US close, so the calm is structural, not just stylistic. Market open/closed (US session state) and data freshness render as small static peripheral indicators[89]. **Probability visuals are motionless — the election-needle lesson. Unchanged.**
- **Position and length over angle and area:** bars, lines, sparklines, small multiples everywhere; gauges, donuts, and speedometers nowhere[77].
- **Dark mode is Desk-only** (optional, for evening use); the Academy stays light — long-form reading favors positive polarity[77].

9.8 What we will not build

Anti-pattern	Why it is banned
Composite buy/sell gauges ("Technical Rating: BUY")	Unauditable aggregates that read as advice — the exact opposite of cause + pattern + base rate.
Ticker tape, flashing cells, jittery dials	Motion manufactures urgency with zero decision value [89].
News firehose / squawk as a primary module	Built for full-time scalpers; for one beginner it is ambient anxiety duplicating movers-with-reason.
Movers without catalyst + RVOL on the row	A bare +21% invites momentum-chasing; "no catalyst + no RVOL = ignore" is the pro rule the UI enforces [92].
P&L as the first thing on screen	Anchors emotion before context [96].
Point predictions or single projected lines	Beginners read a line as a promise [84].
Bare percentages / decimal confidence scores	The chance-of-rain error plus false precision [82].
Win-only track records	Survivorship-filtered history is the core mechanism of trading hype; the log is append-only and complete [94].
Gamification: streaks, XP, confetti, leagues	Regulators linked these mechanics to harmful novice trading frequency [95]; pedagogy is fully separable from them.
Freeform drag-anywhere dashboard builder	Destroys the hierarchy and the zone separation [77].
Black-box scan presets	Hidden criteria turn a preset into an oracle; the visible recipe is the lesson.
Product tours; tooltips on everything; 3+ overlay levels	Forgotten teaching, tooltip blindness, disorientation [78].
Hype copy and anthropomorphic AI voice	Urgency theater and trust inflation; copy stays declarative, neutral, mechanical [77].

Anti-pattern	Why it is banned
Permanent beginner explanations, no fading	Expertise reversal: what helps a novice measurably harms an intermediate [87].
Trending / most-bought discovery	Causally generates loss-making trades (–4.7%/20 days) [52] — see Part 8.

PART 10

Candid uncertainties

A report that demands calibrated honesty from an app owes the same about itself. These are the places where this document's own claims are contested, thinly sourced, or still moving.

- **PEAD's obituary is contested.** "Dead in large caps since ~2006" is the consensus reading, but 2025 papers dispute it — even "settled" decay claims remain live debates[43].
- **The candlestick verdict is method-dependent.** The zero-value result holds under the original exit rules; alternative holding strategies produce contested positive findings[32]. "Contested-negative" is the honest label, and small edges would still be unlikely to survive retail costs.
- **Post-publication decay may be largely a US phenomenon.** No reliable decline appears across 38 non-US markets[11], so decay haircuts calibrated on US data may not transfer elsewhere.
- **The options-implied-move "richness" is small and conditional** — name- and regime-dependent, with the academic straddle literature split[42]. The safe claim is only that no blind buy-or-sell strategy survives retail costs.
- **SPIVA's ~93% is an equal-weighted fund count.** Asset-weighted shortfalls are smaller (~55% by one critique), and Sharpe's arithmetic is a strong approximation rather than a literal identity[5].
- **Daily-direction unpredictability is a regularity, not a theorem.** Small cost-surviving edges existed pre-2010 and may exist in less efficient markets[69]; the protective claim is "no demonstrated persistent retail edge today."
- **A residency correction was applied in revision.** Earlier drafts assumed a Nepal-based user and analyzed that country's outbound-investment restrictions[64]; the user is in fact a US (New York) resident, for whom no such restriction exists. §6.4 now carries the corrected posture — paper-first on purely pedagogical grounds — and source [64] is retained only for citation-numbering stability.
- **Several load-bearing magnitudes are industry data, not peer review:** implied-move beat rates, biotech PDUFA move sizes, and the "most academic strategies fail live" figure[39,49,73]. They are used as context, never as precision inputs.
- **Volatility-persistence figures vary by asset, sample, and estimator.** The settled fact is qualitative — high, durable persistence — not any single autocorrelation number[12].
- **The LLM-headline edge is a live decay experiment.** Whether it has fully decayed is not yet measured; the original authors document decline with adoption[54].

> **WHAT WOULD CHANGE THIS REPORT**
>
> Three findings would force a revision: credible out-of-sample evidence of a cost-surviving retail directional edge; a replication showing volatility forecastability failing at short horizons; or evidence that calibrated base-rate displays worsen rather than improve beginner decisions. Each is monitored by the app's own design — the track-record page is, in effect, a standing experiment on the third.

PART 11

Sources

Numbered as cited. Journal abbreviations: JF = Journal of Finance, JFE = Journal of Financial Economics, RFS = Review of Financial Studies, JBF = Journal of Banking & Finance, QJE = Quarterly Journal of Economics, FAJ = Financial Analysts Journal.

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Research compiled July 2026 by parallel research passes with adversarial cross-examination; twelve load-bearing claims independently challenged and corrected where needed. Provider pricing and platform facts are covered — with per-source verification dates — in the companion volume, *myStockMarket — Build Blueprint*.